AA274A Homework 3

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Collaborators: None
Use of generative AI: Formatting, clarity

Problem 1: Camera Extrinsics from a Checkerboard

(i)

Graded by the autograder; nothing to include.

(ii)

Answer:

$$\mathbf{R} = \begin{bmatrix} -0.8416 & -0.0242 & 0.5395 \\ -0.0465 & -0.9920 & -0.1170 \\ 0.5381 & -0.1236 & 0.8338 \end{bmatrix}, \quad \mathbf{t} = \begin{bmatrix} 0.0604 \\ 0.0550 \\ 0.2484 \end{bmatrix}$$

Reprojection RMSE: 3.303 px; $\|\mathbf{r}_1\| = 1.0277$; $\mathbf{r}_0 \cdot \mathbf{r}_1 = -0.0008$

(iii)

The sign check is necessary because the homography is defined with a scaling between both the x and y axes and the z axis. The transformation could put the checkerboard behind the camera, so we constrain it so that the checkerboard is in front. The closest rotation check is needed because noise produces a matrix that is not orthogonal, so SVD projects it onto the nearest valid rotation matrix.

For a perfect $\mathbf{K}$ and noise-free corners, the re-projected basis vector $\mathbf{r}_1$ should still be of unit length, and it is fairly close. For $\mathbf{r}_0 \cdot \mathbf{r}_1$, since both vectors are supposed to be orthogonal as part of the basis, their dot products should ideally be 0, and it appears that they are very close.

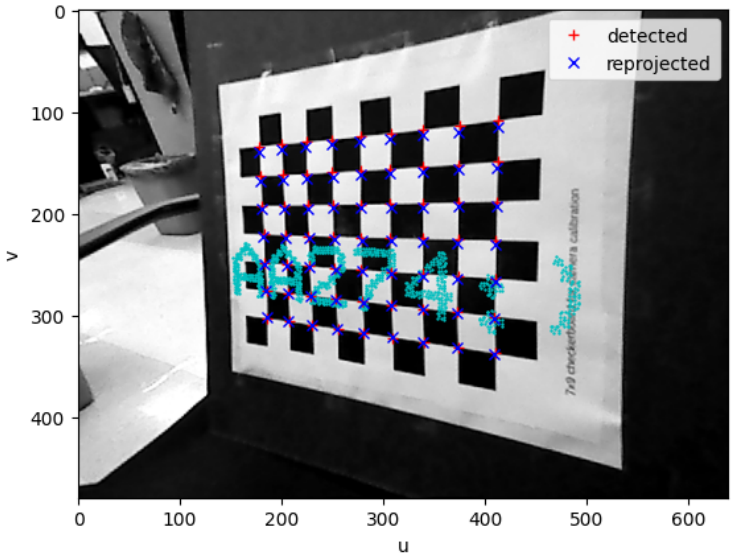
(iv)

The components of $\mathbf{t}$ all change as the scale changes. The third element increases as the scale increases, while the first and second elements decrease. t_z seems to be proportional to the focal length, and that makes sense because the z axis is directly along the axis of the focus. Since the u and v projections are ratios with the axial and transverse coordinates, scaling the focal length scales the other dimensions too.

The rotation change is 0 for the 1.00 scaling, and only a couple of degrees for scaling in either direction. The scale of 1 gives the lowest error, demonstrating that the original K provides the best estimate of the camera intrinsics among these choices.

To estimate $\mathbf{K}$ more reliably, I would use multiple images of the checkerboard taken from different camera positions and orientations.

(v)



Problem 2: Robust Estimation with RANSAC

(i)

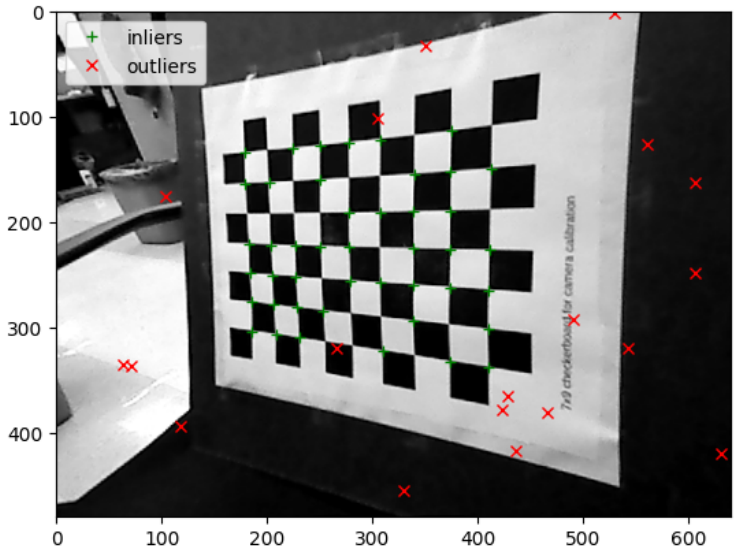
Graded by the autograder; nothing to include.

(ii)

Answer: $s = 4$: $N = 17$; $s = 8$: $N = 78$

N grows quickly with sample size because the probability of having all points as inliers decreases exponentially (since w is always less than 1). Therefore, this requires more iterations for the same confidence.

(iii)



Answer: Least squares rotation error: 93.96°; RANSAC rotation error: 0.273°

RANSAC requires four point correspondences because a homography has eight independent degrees of freedom, and each correspondence provides two constraints. The sampled points must not be collinear because collinear correspondences do not provide enough independent constraints to uniquely determine the homography.

RANSAC repeatedly samples four points and selects the homography supported by the largest number of inliers, making it less sensitive to outliers than least squares, which takes every point into account.

Problem 3: Iterative Closest Point (ICP)

(i)

Graded by the autograder; nothing to include.

(ii)

Table 1: ICP Performance for Different Initial Rotation Offsets

Offset (°)	Iterations	Final NN Dist. (m)	Rot. Error (°)	Point Error (m)
15	100	0.00052	0.05	0.00006
30	100	0.00052	0.05	0.00006
60	100	0.00052	0.05	0.00006
90	100	0.00518	144.68	0.07733
120	100	0.00518	144.84	0.07741

ICP may converge to an incorrect local minimum because a poor initial guess can lead to incorrect nearest-neighbor correspondences, even though the mean nearest-neighbor distance decreases. It seemed to do this for the 90 ° and 120 ° cases. This means the initial guess should be within some angle of the actual transformation.

(iii)

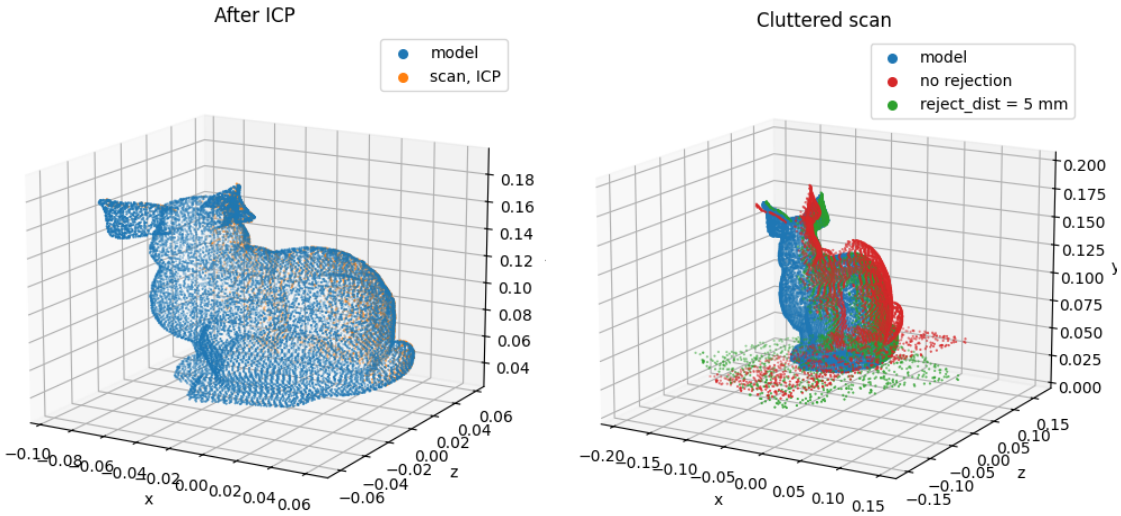
Answer: Without rejection: 9.78°, 0.01246 m; with `reject_dist` = 0.005: 0.27°, 0.00038 m

Floor points bias ICP because they have no corresponding points in the target model while still existing in the source. Without rejection, ICP still matches these points to the nearest model points, adding incorrect correspondences that bias the estimated transformation.

For the final mean distance, the floor points still contribute to the reported distance, even though they are unmatched and unused when making/fitting the transformation.

If the rejection threshold is too small, even valid correspondences are ignored, but a threshold that is too large will include outliers.

(iv)


After ICP
model
scan, ICP
x
z
Cluttered scan
model
no rejection
reject_dist = 5 mm
x
z

Problem 4: Evaluating an Object Detector

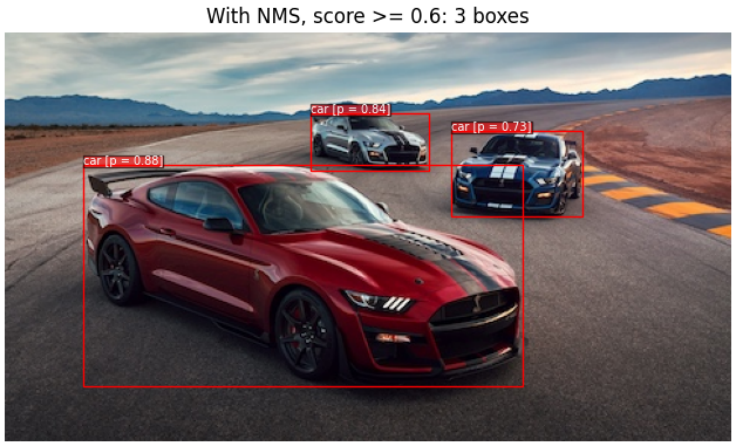
(i)

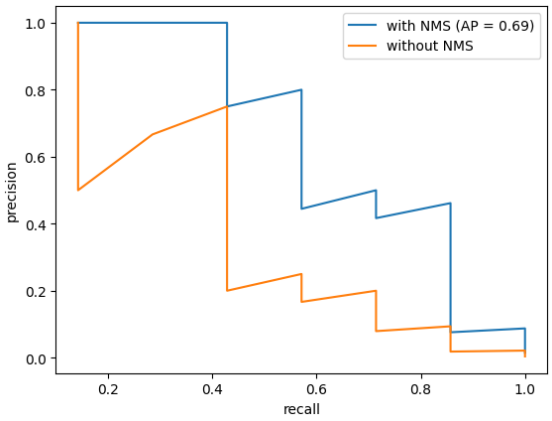
Graded by the autograder; nothing to include.

(ii)

Answer: AP: 0.693; at threshold 0.5, precision: 0.800, recall: 0.571

(iii)

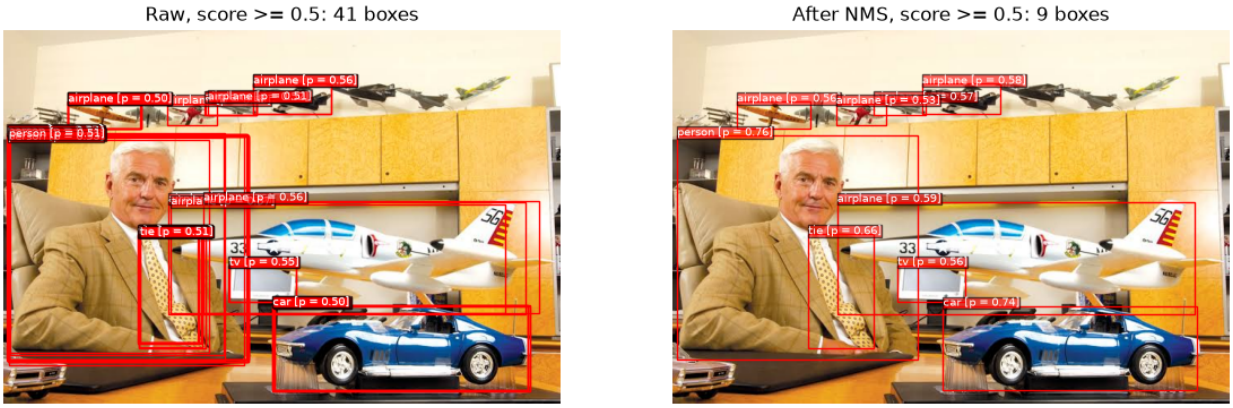




When the threshold is raised to 1, almost all detections are removed because the threshold is so high. This corresponds to the recall going to 0 (not detecting any relevant items, so false negatives), which makes the precision undefined because there are no predictions. When the threshold is lowered to 0.01, it keeps all detections, increasing the recall, but also means that many of the items retrieved will be irrelevant (false positives).

Without NMS, multiple detections of the same object can be counted as false positives after the first successful match, reducing precision and therefore AP.

(iv)



The model did a really good job of detecting the various types of objects in the scene, such as the planes, car, tv, and tie, even with the TV partially obstructed in the background. It did miss some planes on top of the cabinets though.

The image must be converted to floating-point values in $[0, 1]$ because FCOS ResNet-50 FPN expects normalized image values, and it must be passed as a list because torchvision expects a batch of image tensors.